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import java.util.Scanner;

public class MergeSort {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Get array size from user
        System.out.print("Enter number of elements: ");
        int n = scanner.nextInt();

        // Create and initialize an array
        int[] arr = new int[n];
        System.out.println("\nEnter element values:");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        // Sort the array using Merge Sort
        mergeSort(arr, 0, n - 1);

        // Print the sorted array
        System.out.println("\nSorted array:");
        for (int i : arr) {
            System.out.print(i + " ");
        }
        System.out.println();
    }

    private static void mergeSort(int[] arr, int low, int high) {
        if (low < high) {
            // Find the middle point
            int mid = (low + high) / 2;

            // Sort the left and right halves recursively
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        mergeSort(arr, low, mid);
        mergeSort(arr, mid + 1, high);

        // Merge the sorted halves
        merge(arr, low, mid, high);
    }
}

private static void merge(int[] arr, int low, int mid, int high) {
    // Create temporary arrays for left and right sub-arrays
    int leftSize = mid - low + 1;
    int rightSize = high - mid;
    int[] left = new int[leftSize];
    int[] right = new int[rightSize];

    // Copy data to temporary arrays
    System.arraycopy(arr, low, left, 0, leftSize);
    System.arraycopy(arr, mid + 1, right, 0, rightSize);

    // Merge the temporary arrays back into arr
    int i = 0, j = 0, k = low;
    while (i < leftSize && j < rightSize) {
        if (left[i] <= right[j]) {
            arr[k] = left[i];
            i++;
        } else {
            arr[k] = right[j];
            j++;
        }
        k++;
    }

    // Copy remaining elements (if any)
    while (i < leftSize) {
        arr[k] = left[i];

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        i++;  
        k++;  
    }  
    while (j < rightSize) {  
        arr[k] = right[j];  
        j++;  
        k++;  
    }  
}  
}
```